

Average Value

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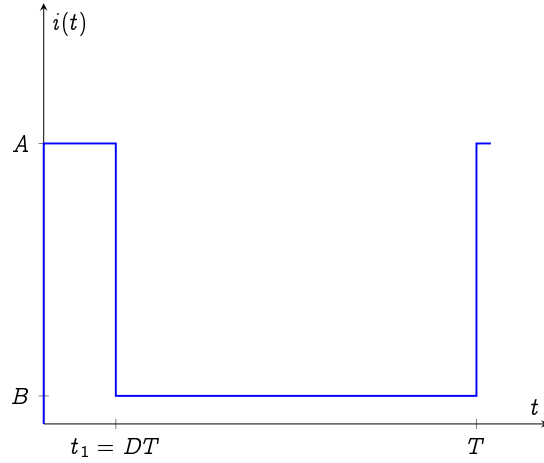


Figura 1:

$$i(t) = \begin{cases} A, & 0 \leq t \leq t_1 \\ B, & t_1 \leq t \leq T \end{cases} \quad (1)$$

The average value is given by:

$$I_{avg} = \frac{1}{T} \int_0^T i(t) dt \quad (2)$$

substituting (1) into equation (2) we get:

$$\begin{aligned} I_{avg} &= \frac{1}{T} \left[\int_0^{t_1} A dt + \int_{t_1}^T B dt \right] = \frac{1}{T} \left[At \Big|_0^{t_1} + Bt \Big|_{t_1}^T \right] \\ &= \frac{1}{T} [At_1 + B(T - t_1)] \end{aligned} \quad (3)$$

If we assume that $t_1 = DT$ where $D = \frac{t_1}{T}$ then:

$$I_{avg} = \frac{1}{T} [ADT + B(T - DT)] \quad (4)$$

$$I_{avg} = AD + B(1 - D) \quad (5)$$

If $A = 20 \text{ mA}$, $B = 60 \text{ } \mu\text{A}$, $t_1 = 20 \text{ } \mu\text{s}$, $T = 10 \text{ ms}$, then

$$D = \frac{t_1}{T} = \frac{20 \text{ } \mu\text{s}}{10 \text{ ms}} = 2 \times 10^{-3} \quad (6)$$

$$I_{avg} = (20 \text{ mA})(2 \times 10^{-3}) + (60 \text{ } \mu\text{A})(1 - 2 \times 10^{-3}) = 99.88 \text{ } \mu\text{A} \quad (7)$$